

Explainable Longitudinal Digital Twin for Glioma Progression: A Real-Data Multimodal MRI and Clinical Approach

Abstract

The monitoring of Glioma progressive is still a difficult clinical practice because of the complexity of the temporal changes of tumor characteristics and the availability of magnetic resonance imaging (MRI) analysis at a single timepoint[1-4]. While Deep learning (DL) methods have shown good performance in the field of brain tumor classification and segmentation, most researches in the field have considered brain tumor classification problems with single or static MRI images and ignored the changes that have happened between two consecutive clinical visits[2-9]. Moreover, the interpretation of deep learning systems in medical images is still a major issue, especially if the learned attention patterns are determined to be clinically relevant tumor evolution[10-15]. In this paper, we introduce a novel method that combines a Siamese Convolutional Neural Network (CNN), attention-based slice aggregation, and the fusion of clinical features to create an explainable longitudinal learning framework for glioma progression prediction[6-16]. The proposed framework learns the temporal representations from paired MRI exams, sampled at a first and a second timepoint, by (1) extracting feature embeddings (first timepoint), and (2) modeling the difference between longitudinal representations (seconds timepoint)[6][8][9]. A longitudinal glioma cohort of 298 patients and 1,788 MRI scans was used. For each patient a $48 \times 2 \times 224 \times 224$ is presented as the Anatomical Slices (AS) for each of two MRI modalities (FLAIR and T1CE). The proposed framework was compared against various baseline models such as clinical-only prediction model, CNN-based single-timepoint prediction model and conventional Siamese MRI prediction model. The experimental results have shown that the attention enhanced Siamese model for MRI by incorporating clinical features obtained the best performance among all other models, with accuracy of 60.00% and Macro-F1 score of 0.4006 with validation. An untouched test data set, however, saw lower generalization performance, with the accuracy of 46.67% and the Macro-F1 score of 0.3051 compared to the clinical-only baseline at 73.33% and 0.6616. Moreover, while using the multiple variants of Grad-CAM for explainability analysis, the authors found little spatial overlap between model focus and expert-drawn tumor regions, which had Dice overlap values around 0.01 [10-13][15][17][18]. These results underscore the need for assuming medical-quality AI systems not only have predictive accuracy but are also reliable when they explain their predictions[10-14][17][18]. The framework proposed allows for a thorough investigation on predicting gliomas from the disease's longitudinal MRI sequences through the methods of

temporal representation learning, multimodal feature fusion, and quantitative explainability[6][8][9][16].

Index Terms— Glioma progression, longitudinal MRI, Siamese convolutional neural network, attention mechanism, explainable artificial intelligence, Grad-CAM, medical deep learning.

I. INTRODUCTION AND RELATED WORK

Gliomas are one of the most aggressive primary brain tumors which continue to have high morbidity and mortality even with the recent improvement in diagnosis and treatment. To provide personalized treatment planning, patient monitoring, and clinical decision support, it is crucial to make an accurate prediction of the tumor's progression. Because of the detailed structural and functional information it can provide about the characteristics of a tumour, magnetic resonance imaging (MRI) has become the main imaging technique used in the assessment of gliomas. The prediction of future tumor behavior from MRI is still difficult, however, due to glioma evolution being a dynamic process with complex spatial and temporal variations [19][20].

The traditional clinical measurement of glioma progression is based on the radiological interpretation of sequential MRI exams. Evaluation by experts can help, but there is inter-observer variability and not all experts can detect subtle longitudinal changes. Thus, there has been a growing interest in machine learning and deep learning methods to automatically learn predictive representations from medical image data [21][22].

In medical image analysis, convolutional neural network (CNN) has been used with great success in various applications, such as tumor segmentation, classification, and survival prediction. CNN-based methods are able to automatically learn hierarchical imaging features without the need of handcrafted features. Several studies have shown that deep neural networks can detect patterns from MRI images that relate to the presence of tumors and perform as well as traditional machine learning techniques [21][23].

Most of the currently used MRI-based glioma prediction techniques, however, present a prediction for each individual scan. These approaches treat MRI examinations as snapshots of time and fail to take into account the time-ordered nature of subsequent patient visits. When applied clinically, the progression is not just the appearance of the one MRI scan – it is the changes over time. Thus, the modelling of tumor evolution over time requires longitudinal models [24].

Siamese neural networks have become successful for learning similarities and differences among paired inputs. Siamese architectures can share common network parameters to produce similar feature representations of two related observations. This makes them useful in longitudinal medical imaging applications, where the goal is to understand what is going on in the imaging data between different time points. For glioma analysis, Siamese models have been used for tumor change detection and modelling its progression, but most of the literature only addresses image level comparison and does not consider other clinical information [25].

Current deep learning-based medical systems have another drawback: Interpretability. Deep models can attain a high predictive performance but little is known to clinicians about their decision-making process. To visualize areas that contribute to model predictions, explainable artificial intelligence (XAI) techniques have been widely used, including Gradient-weighted Class Activation Mapping (Grad-CAM) [26]. However, clinical validity for visual explanation is not guaranteed. It is still a research challenge to determine if there is a quantitative relationship between model attention and real pathological regions [27].

Recently, attention mechanisms have been added to medical imaging models to enable the network to selectively concentrate on informative regions or instances. In longitudinal MRI analysis, attention-based aggregation can help the model to focus on the relevant tumor slices and ignore all other spatial slices [28]. Learned attention weights do, however, need systematic exploration to see if they really reflect tumor evolution.

Other clinical parameters like patient characteristics and diagnosis information might also give additional information besides the imaging features. Several multimodal learning frameworks that integrate MRI representations with clinical information have been proven to enhance prediction accuracy [29]. The relationship between temporal features derived from imaging data and clinical data in longitudinal glioma prediction, however, has not been sufficiently explored. To overcome these shortcomings, the authors propose a longitudinal framework for predicting gliomas that imposes online explainability requirements by introducing a Siamese MRI representation learning module, an attention-based slice pooling module, and a module for clinical feature fusion. While conventional single-timepoint methods fail to explicitly model MRI changes between visits, the proposed framework explicitly models MRI changes between visits and evaluates the reliability of the learned attention with quantitative explainability analysis.

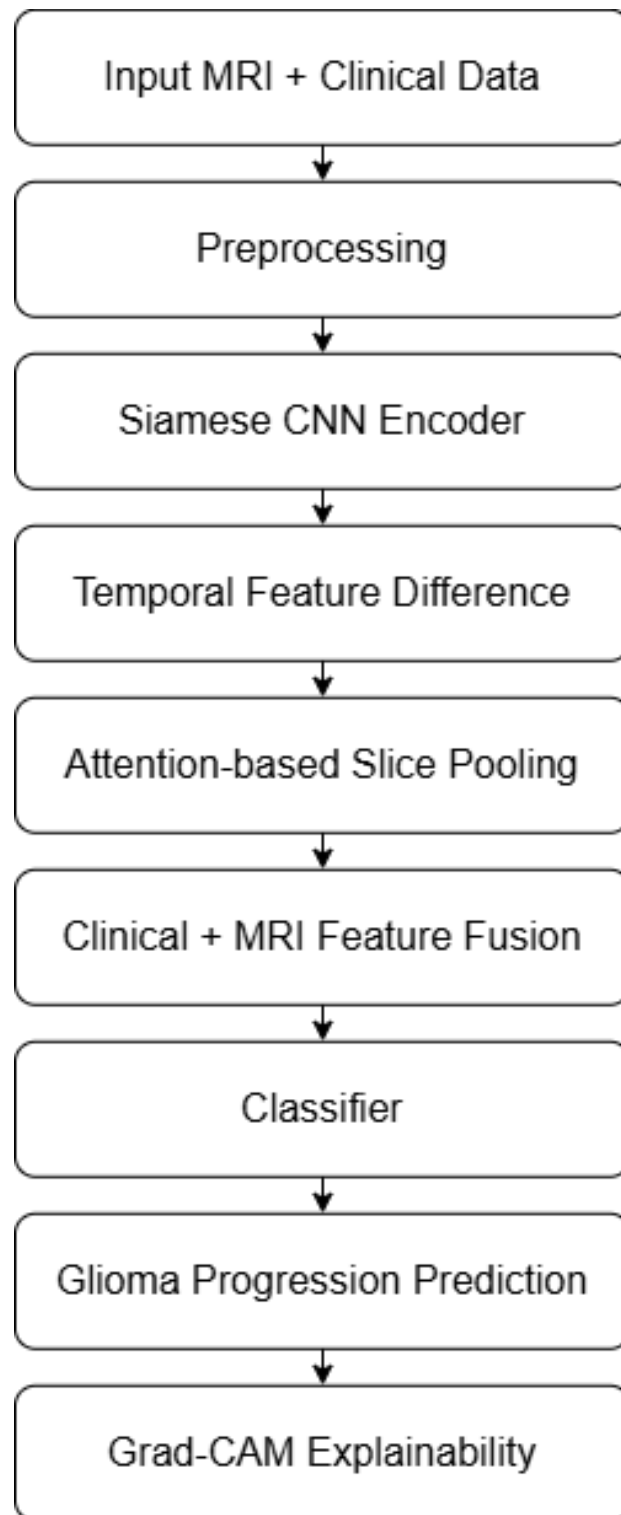


Fig 1: Overall framework architecture

II. METHODOLOGY

A. Dataset

The present study was performed on a longitudinal cohort of gliomas patients (298 patients with two acquisitions of MRI scans at different timepoints). A total of 1788 MRI files were assessed, and all cases were checked for general completeness/reliability of the data.

For each patient, a baseline and follow-up MRI scan and respective tumor segmentation data with patient-specific clinical information was provided. Clinical characteristics included age, sex and diagnosis data. Tumor label/Enhancing tumor was extracted and sharpened in the enhancing tumor (ET) region from experts' mask segmentation results.

In order to avoid leakage of information for tumor progression, the information related to the difference in Segmentation was not included as an input feature. The data set was split on a patient basis with 208 cases used for training, 45 for validation, and 45 untreated cases for final (test) evaluation.

B. MRI Pre-processing and input representation

The MRI exam results from various patients can have differences in intensity distribution, spatial resolution and positioning of anatomical structures. As such, a standardized preprocessing pipeline was used and consistent model inputs produced.[30], [31]

Firstly, variations among different scanners were minimized by intensity normalization.

$$X_{\text{norm}} = (X - \mu) / \sigma$$

where XX represents the original MRI intensity; μ is the average of the intensities of the MRI; σ is the standard deviation of the intensities of the MRI.

The MRI volumes were then cropped, padded, and resized to a uniform spatial resolution of $224 \times 224 \times 224$. From both MRI scans, axial slices that labeled each corresponding structure were chosen so that longitudinal axis would provide a correspondence.

Each patient came out at the end as:

$$X \in \mathbb{R}^{48 \times 2 \times 224 \times 224}$$

The selected axial slices are given by the value 48, the MRI channels (FLAIR / T1CE) can be selected by the value 2, and $224 \times 224 \times 224$ indicates the spatial resolution.

This standard representation allowed a standard comparison between baseline and follow up MRI examinations.

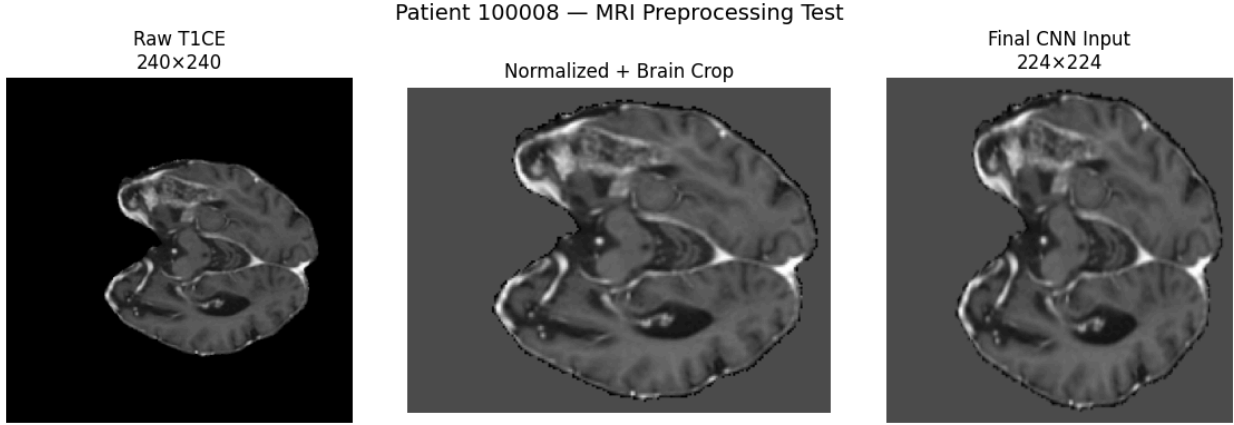


Fig 2: MRI preprocessing pipeline and final representation in the longitudinal domain.

C. Longitudinal Siamese CNN Framework

The framework was created to learn tumour evolution, based on longitudinal MRI images. To extract end-to-end similar characteristics from paired inputs, the proposed Siamese network extracts features for two MRI scans (from the same brain) by sharing a common feature extraction network with the CNN.

If there are two MRI exams, inputs are provided as a pair of variables: X_1 and X_2 . X_1 is the MRI exam performed at Time 1, and X_2 is the MRI exam performed at Time 2. The two inputs are fed into the same CNN's encoder:

$$F_1 = f_\theta(X_1)$$

$$F_2 = f_\theta(X_2)$$

The extracted feature embeddings are F_1 and F_2 , where a shared CNN encoder is denoted as f_θ .

The difference between the times of two examinations is calculated next:

$$\Delta F = F_2 - F_1$$

By using such a time-based representation, the model is not required to learn the static pattern of the tumors, but may just focus on finding the changes between the exams.

D. Attention-Based Slice Aggregation

One of the challenges in deep learning with MRI is the fact that every slice of the MRI does not have the same amount of informative data for diagnosis. Common methods of pooling assume homogeneity, but may yield sub-optimal contributions from clinically informative regions.

Towards handling these problems, an attention-based slice aggregation mechanism was introduced. The model computes an importance value for each slice and decides on those adaptive weights based on them.

The attention weight is calculated as:

$$\alpha_i = e^{s_i} / \sum_j e^{s_j}$$

where s_i is the slice's importance score, slice i is the current slice.

Finally, the MRI is given at the patient level as follows:

$$Z = \sum \alpha_i F_i$$

where F_i is a feature representation of its corresponding slice.

This can be used as a mechanism for highlighting informative slices pertaining to tumour information and minimizing the impact of non-informative areas.

E. Risk and Resilience to Accomplishment

While MRI is useful for the spatial and temporal information, patient-level characteristics can provide complementary information. Thus the longitudinal MR image has been added with clinical features, to get the extracted image.

The feature representation for the multimodal was achieved by:

$$H=[Z,C]$$

where Z denotes feature representation from MRI and CC denotes clinical feature vector.

The fused representation was then processed by fully connected layers to get the final progression classification.

F. Representative Patient in Digital Twins.

A patient specific latent representation was created from information obtained from the fused longitudinal MRI.

The computational patient state was given as:

$$D_i = g(H_i)$$

Where D_i denotes the learnt latent state of patient i .

This is based on the implementation of Digital Twin, with computational model holding imaging and clinical features for an individual patient. But this should be thought of as a 'Digital Twin' inspired model representation and not as a complete biological 'Digital Twin'.

G. Explainability and Tumor Evolution Validation

Explainability analysis based on visualization methods from the Grad-CAM have been conducted to explore the model's learning of clinically meaningful patterns.

The basic form of Grad-CAM is:

$$L^c = \text{ReLU}(\sum \alpha_k^c A_k)$$

where A_k represents feature activation maps and α_k^c represents their importance weights.

Standard Grad-CAM subtracts negative parts using ReLU location, thus additionally, Signed Grad-CAM was assessed:

$$L_{sc} = \sum \alpha_k A_k$$

These attention maps were quantitatively compared with maps identified by the experts in the enhancing tumor regions using dice similarity:

$$\text{Dice} = (2|A \cap B|) / (|A| + |B|)$$

The areas highlighted are the regions of interest in the model (A) and the tumor masks from the expert (B).

In addition, the temporal alignment analysis method was applied to examine if the update of the model-attention was correlated to the real tumors changing during the experiment and the centroid changing analysis method was used for checking the statistical significance of the results.

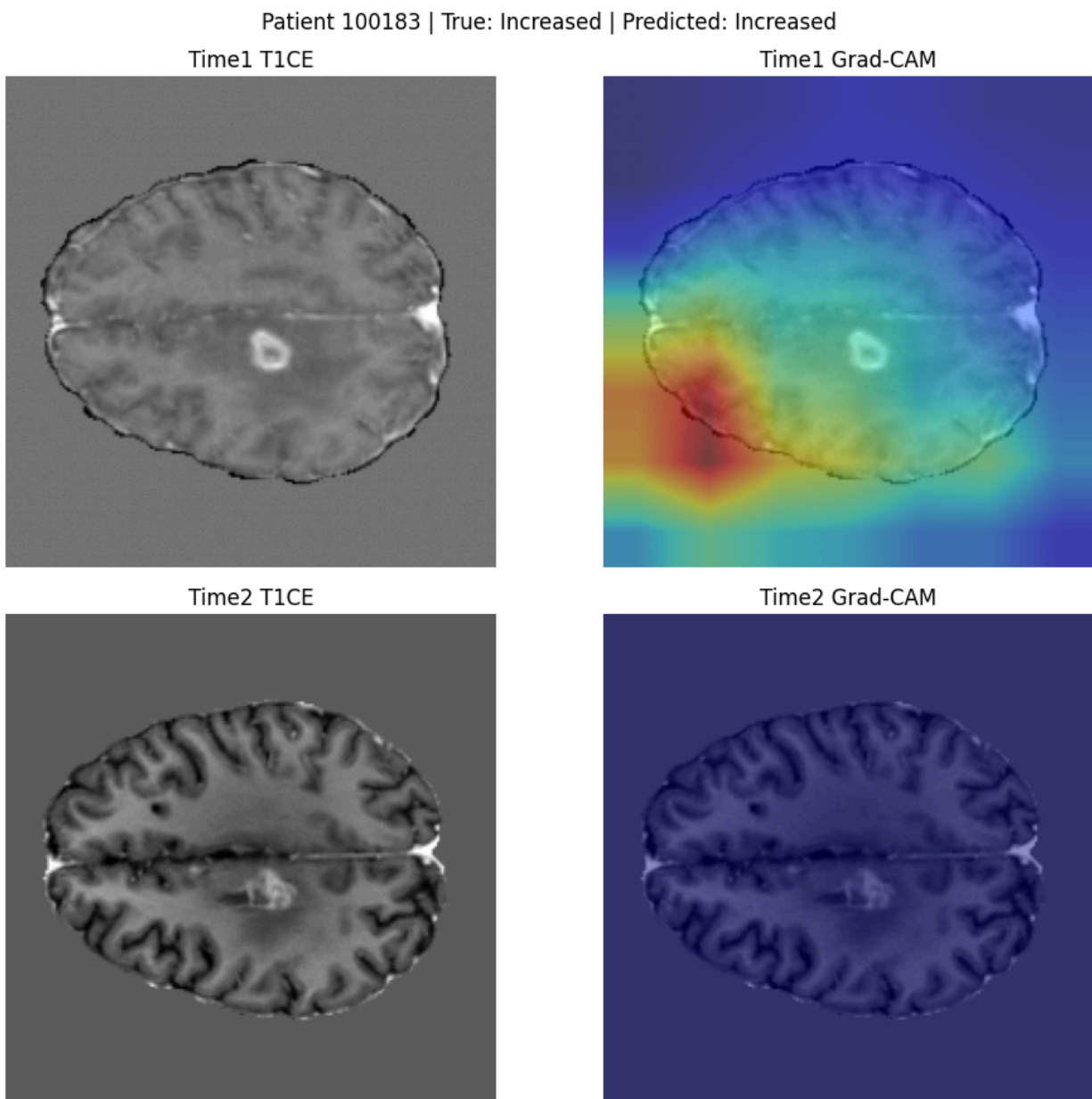


Fig 3: The main goal of this goal is to explain the validation framework that compares the model attention to tumor evolution.

H. Model Training and Evaluation

A progressive experimental approach was used to evaluate the proposed framework. Four models were developed for each analysis—clinical-only baseline, CNN models with a single timepoint, Siamese MRI only model, Siamese model with clinical fusion, and the final attention-enhanced Siamese model.

The accuracy and Macro-F1 score are used for evaluation:

$$\text{Accuracy} = \text{Correct Predictions} / \text{Total Samples}$$

$$\text{Macro - F1} = (1/K) \sum F1_i$$

Further, statistical comparisons between the models were made using both bootstrap confidence intervals and McNemar's exact test.

III. RESULTS AND DISCUSSION

A. Experimental Results and Model Comparison

A series of experiments in a variety of combinations of imaging data and clinical data was performed to assess the performance of the longitudinal glioma progression prediction approach. The experiments were aimed at exploring the role of longitudinal MRI representation, clinical feature integration, and feature aggregation based on attention.

First, various baseline models were tested such models based on clinical data only and CNN models based on one time-point MRI. Then, the longitudinal models were constructed using pairs of MRI scans done on the same patient, based on Siamese architectures.

All the models are summarized in the validation performance in Table below.

Table 1 : Validation performance

Model	Accuray	Macro-F1
Clinical-only	0.4444	0.400
zTime1 CNN	0.5333	0.2676
Time2 CNN	0.5333	0.3251
Siamese MRI-only	0.4444	0.2894
Siamese + Clinical	0.4222	0.2928
Improved Siamese + Clinical	0.6000	0.4006

The clinical-only baseline showed an accuracy of 0.4444, highlighting that basic demographic and diagnostic data is only a partial signal of glioma progression.

In terms of accuracy, the accuracy values of MRI were 0.5333 with single-timepoint MR images (Time1 and Time2), while those of the CT images were 0.3750. These findings showed that the

MRI contains advanced image characteristics related to tumor status. They did not perform as well, however, on their Macro-F1 scores, which left questions about their ability to provide a balanced classification in all categories of progression.

In order to obtain the information of time, it has become necessary to make a comparison between two MRI examinations; hence the Siamese MRI-only model has been developed. Its performance, however, was not as good as single-timepoint CNN models, suggesting that learning temporal features with a few usually short timespan observations is still a difficult task.

The model's performance was comparable when the clinical features were combined with the Siamese MRI representation. This indicates that multimodal fusion may be unable to combine imaging and clinical information effectively with simple multimodal fusion.

The proposed Improved Siamese + Clinical model enhanced the performance of the validation task, obtaining the best accuracy. The model achieved:

$$\text{Accuracy} = 0.6000$$

$$\text{Macro-F1} = 0.4006$$

The improvement suggests that the aggregation to control attention was useful for the network to learn more relevant MRI slices, and for it to produce more discriminative patient embeddings.

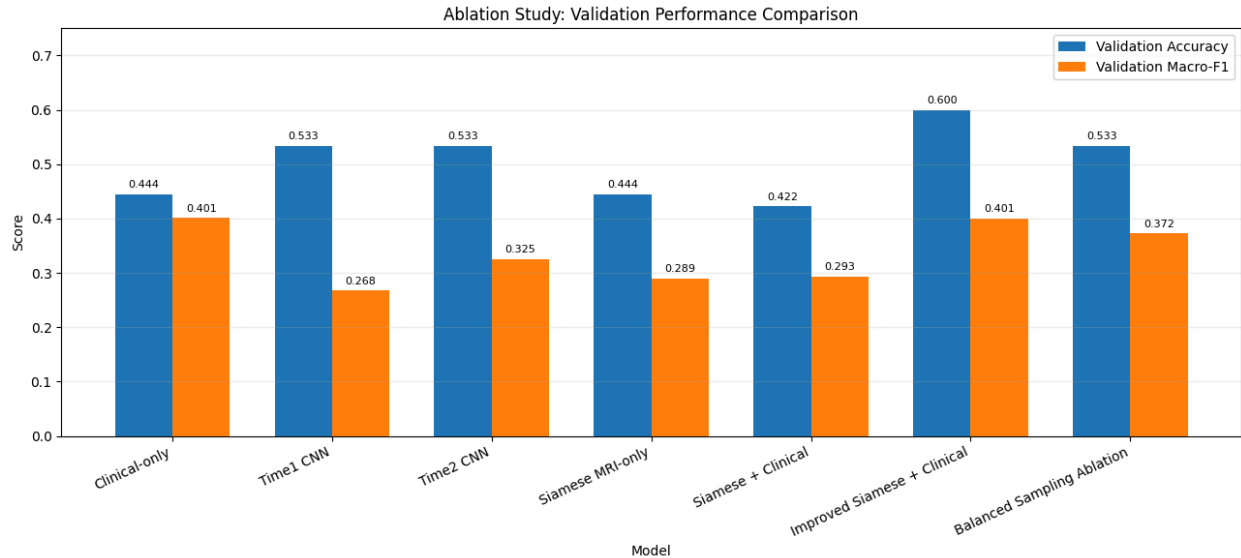


Fig 4: Performance comparison of models with regards to validation.

B. Effect of Attention-Based Slice Aggregation

In order to better represent the volumetric MRI data, an attention-based aggregation method is used in place of the conventional slice pooling method.

In MRI, there are a large number of slices, however only a limited number of these slices will have tumor related information. Averaging methods assume that each slice contributes equally, and can minimize the effects of clinically relevant areas.

The attention mechanism can give different attention weights to the different slices, and dynamically sum up the information.

The validation accuracy of the Siamese + Clinical model raised from 0.4222 to 0.6000 with the introduction of attention based pooling mechanism in the Improved Siamese + Clinical model.

This is the evidence of the better performance offered by the adaptive feature aggregation approach over uniform slice contribution at patient level.

C. Balanced Sampling Analysis.

Since progression categories showed class imbalance, a balanced sampling strategy was additionally evaluated.

They got the following balanced sampling model:

Accuracy: 0.5333

Macro-F1: 0.3723

Balanced sampling did result in improved representation of each class, but did not outperform the attention-based model.

Thus, the Siamese + Clinical model with focus enhancement was chosen to be evaluated independently using MRI.

D, Independent Test Evaluation.

To assess for the ability to generalize the selected model was tested on an untouched test cohort of 45 patients.

This cohort received no further training, optimization of parameters or selection of the model.

The final test performance is given in Table II.

TABLE 2 : Performance comparison on the independent test cohort

Model	Accuracy	Macro-F1
Clinical-only baseline	0.7333	0.6616
Improved Siamese + Clinical	0.4667	0.3051

In an independent analysis, an important distinction between validation performance and generalization in the wild was noticed.

The proposed MRI-based model obtained the best validation performance; however, the clinical-only baseline outperformed in cases of unseen patients.

The model performed clinical only achieved:

Accuracy: 73.33%

Macro-F1: 66.16%

While the last multimodal MRI model was able to achieve:

Accuracy: 46.67%

Macro-F1: 30.51%

The discovery suggests that the MRI-based model learned informatively during development, but that the complexity of unseen test patients is too great to support as well.

Possible explanations include:

1. Small number of patients per cohort:

The images need to be represented with high level in order to be represented by deep learning models; hence large scale data sets are required.

2. High-dimensional MRI characteristics:

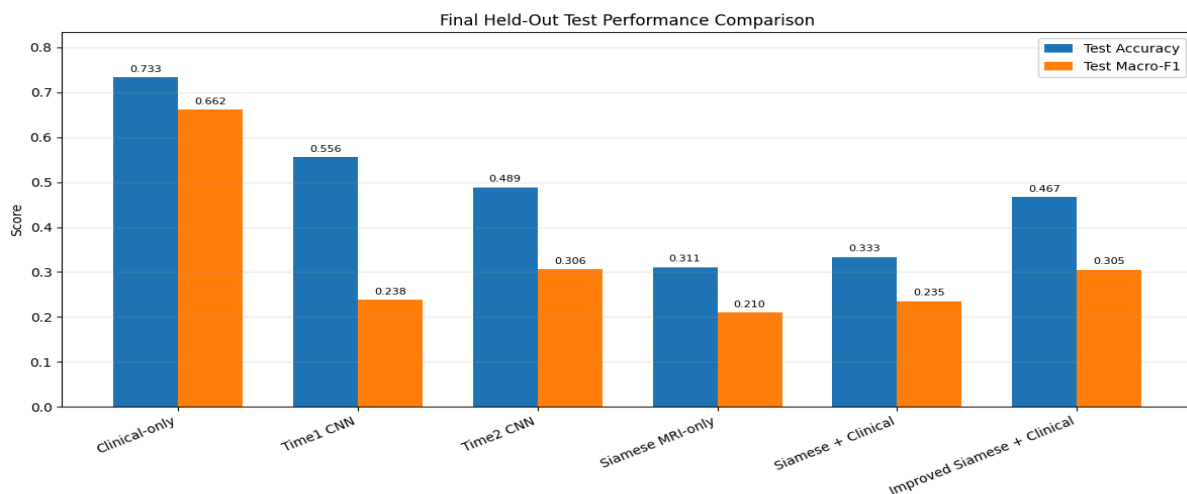
MRI has a lot of spatial information which may lead to learning dataset specific patterns.

3. Tumor heterogeneity:

Progression of gliomas is different for each individual.

4. Limited longitudinal observations:

However, only two MRI timepoints were available, so this may not reflect tumor evolution over time.



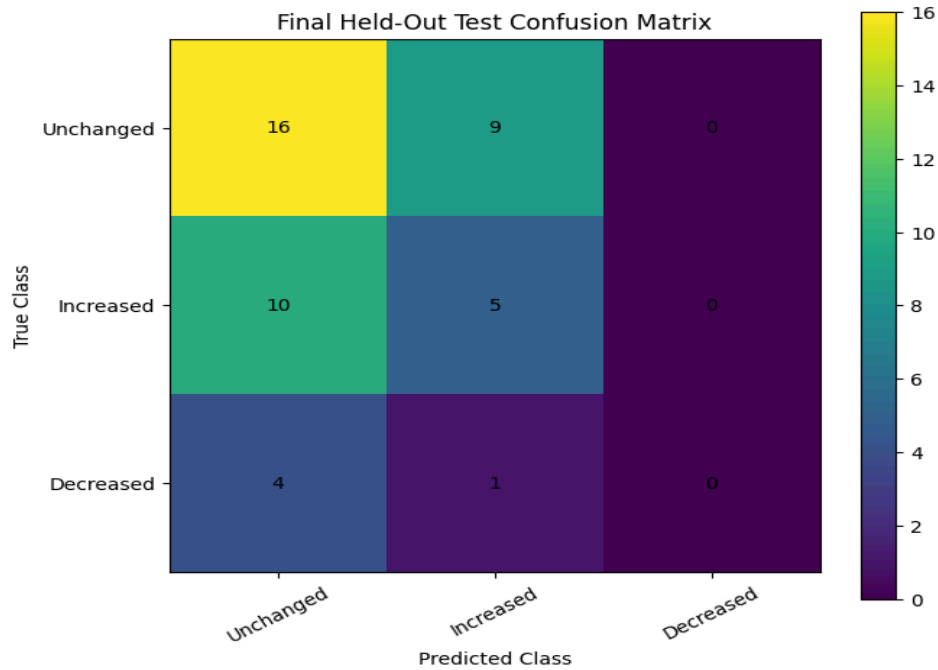


Fig 5: Compare the result of the final tests, and construct a confusion matrix. Compare the final test performance and create a confusion matrix.

E. Statistical Significance Analysis

McNemar's exact test was used to compare the prediction differences of two models: clinical-only and final multimodal.

The obtained p-value was:

$$p=0.028959$$

According to the result, it shows that the Two Model is statistically significant with respect to the paired prediction behavior.

This demonstrates that the clinical-only and MRI-based models acquired different subsampling decisions on the same test set.

F. Explainability Analysis

1) Grad-CAM Stability Evaluation

A clinically meaningful region analysis-Forty explanations by Grad-CAM was conducted to test the extent to which the model learnt clinically meaningful regions.

Firstly, conventional Grad-CAM, drew only blank maps of certain inputs,

Table 3 : Conventional Grad-CAM

MRI Timepoint	Zero CAM
Time1	16/45
Time2	8/45

This is due to the fact that standard Grad-CAM uses ReLU activation that removes any negative feature contributions.

So, the performance of the Signed Grad-CAM method was tested.

After applying Signed Grad-CAM:

Table 4 : Signed Grad-CAM

MRI Timepoint	Zero CAM
Time1	16/45
Time2	8/45

This demonstrates improved explanation stability.

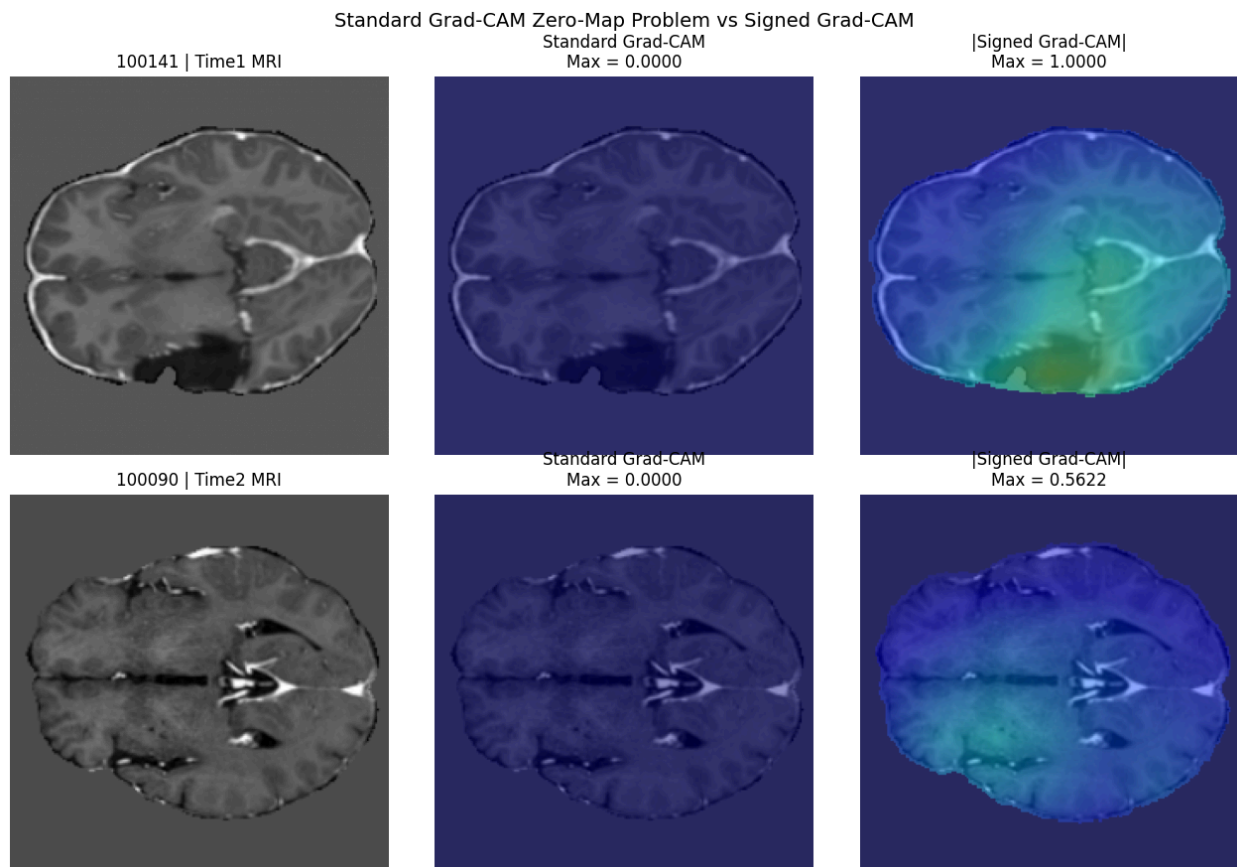


Fig 6: Conventional Grad-CAM vs Signed Grad-CAM visualization.

G. Quantitative Validation of Model Attention

Visual assessment of AI explanations alone is not enough to prove that they match the biological regions of a tumor. So, quantitative comparison of attention maps and the masks manually defined by experts for enhancing tumors was performed.

The results of spatial overlap are summarized in Table below.

TABLE 5 : Attention and tumor region overlap analysis

Method	Time1 Dice	Time2 Dice
Single-layer Grad-CAM	0.0100	0.0106
Multi-layer Grad-CAM	0.0110	0.0119

The Dice overlap was very low and the presence of multiple layers in the Grad-CAM did not help much.

The areas highlighted by the model are not well correlated with the expert tumor segmentation areas, which indicates that this is not a good model performed across all regions.

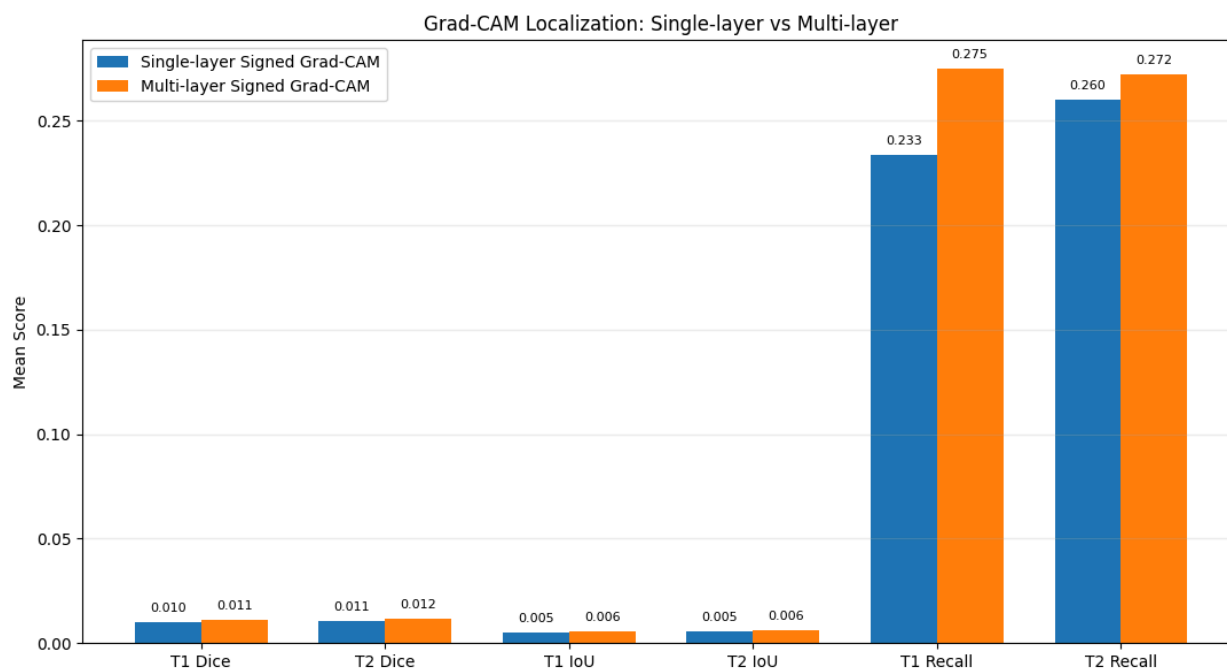


Fig 7: Spatial comparison between model attention maps and expert tumor masks.

H. Temporal Attention Alignment Analysis

Another hypothesis tested was whether model attention shifted as a result of real tumor evolution.

A comparison was undertaken between the temporal alignment of:

- i) Tumor evolution
- ii) Attention evolution

The results obtained are presented in the following Table IV:

TABLE 6 : Temporal attention alignment analysis

Analysis	Result
Single-layer alignment score	0.0039
Single-layer Wilcoxon p-value	0.809166
Multi-layer alignment score	-0.0012
Multi-layer Wilcoxon p-value	0.854390

As a result of the Wilcoxon test there was no statistically significant temporal alignment:

$$P > 0.05$$

As a result, model attention did not correspond to the actual progression of tumors.

I. Tumor Shift and Attention Shift Relationship

To explore this further, centroid displacement analysis was conducted to determine if attention movement took place with the tumor motion.

According to the results of the correlation, the following correlations were:

Pearson correlation:

$$R = 0.0549, P = 0.777248$$

Spearman correlation:

$$R = -0.1133, P = 0.558418$$

The weak correlation values show that there was no significant correlation between the model attention displacement and the tumor displacement.

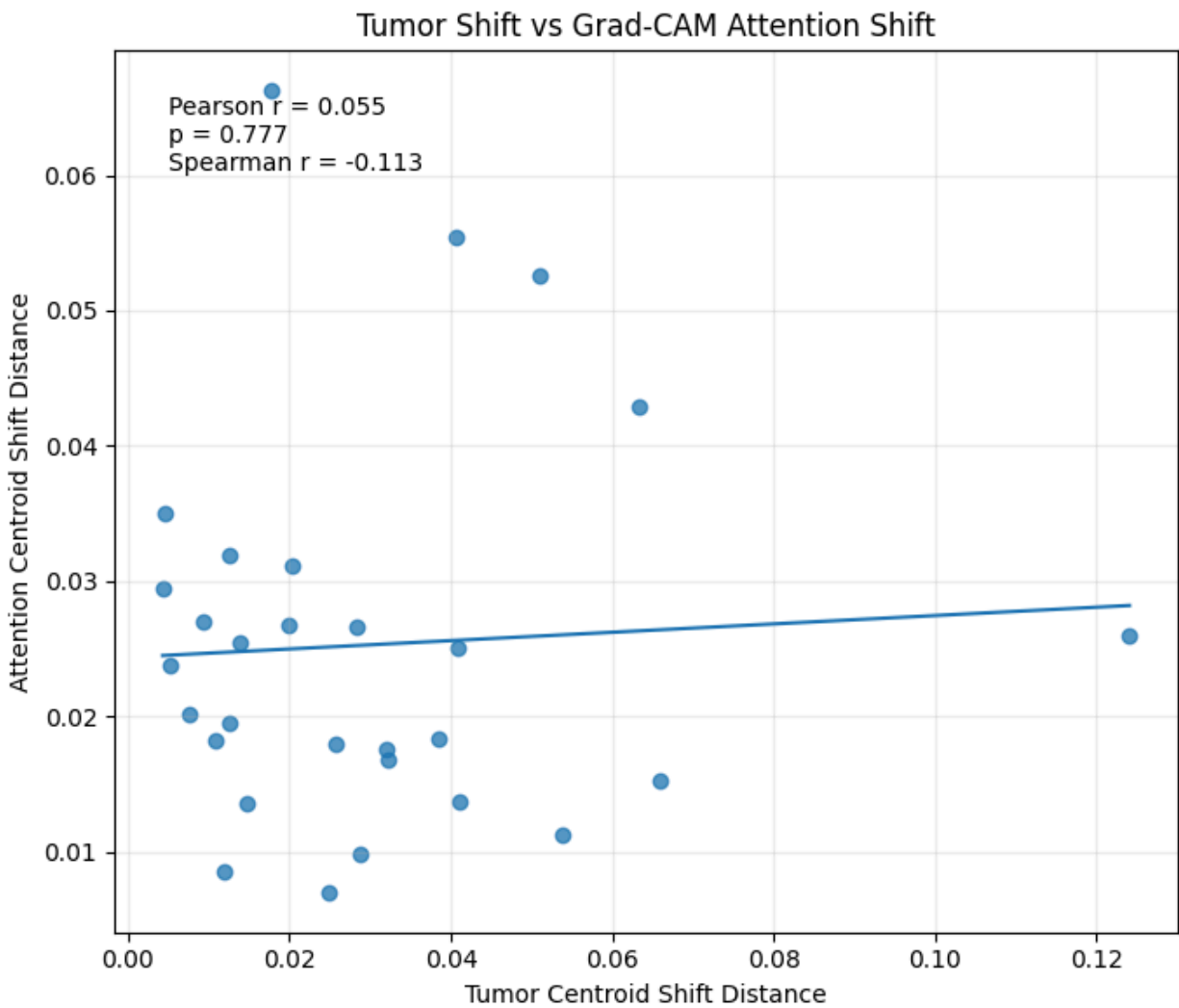


Fig 8: The relation between the shift in the centroid of the tumour and the shift in the centroid of the attention.

IV. CONCLUSION

This study introduced an explainable longitudinal deep learning model for predicting glioma progression, which combines multi-timepoint MRI analysis, temporal feature learning via Siamese CNN, and attention-based slicestack aggregation and clinical feature fusion. The proposed framework attempted to learn tumor evolution by learning the differences between baseline and follow-up MRI scans in contrast to traditional methods which considered individual MRI exams in isolation.

It was tested on a longitudinal cohort of 298 glioma patients. Various experiments were performed to assess the effects of various components (clinical information, single-timepoint MRI features, Siamese representation over time and attention-based feature aggregation). The model with Improved Siamese + Attention was the best-performing model among the validation models, with a score of 0.6000 and an F1 score of 0.4006.

When tested on the independent test cohort however, clinical-only baseline performed better in term of generalization. The clinical-only model has achieved 0.7333 accuracy and 0.6616 Macro-F1 while final longitudinal MRI model gets 0.4667 accuracy and 0.3051 Macro-F1. This discovery underscores the difficulty in creating comprehensive deep learning models for medical imaging applications, such as those involving small sample sizes from longitudinal studies and large variations in tumor types.

In addition to the prediction performance, this work explored whether the model explanations learnt showed agreement with actual tumor evolution. While we eliminated explanations that were zero valued in the conventional Grad-CAM by computing Signed Grad-CAM, quantitative comparisons showed that the activated area of the model and the area of interest defined by experts did not well overlap. In addition, temporal attention analysis revealed no significant correlation between the trend of attention evolution and the actual tumor progression trend.

This study's greatest merits are the systematic assessment of predictive and reliability of explanation in GLIA. The results show that if one judges the results of deep learning models by only their performance, one cannot assume that they can be trusted in clinical practice without performing quantitative checks on their reasoning.

A larger multi-center longitudinal data, inclusion of additional time points of MRI scans, more advanced 3D architectures and transformer-based architecture, and more biologically informed, explainable-AI approaches will be explored in future research and development to enable successful generalization; more reliable clinical decision-support will be achieved.

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